

DEV KUMAR

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PROFESSIONAL SUMMARY

Software Engineer (Principal Core Infrastructure | Java, GoLang, Python | Agentic AI & Distributed Systems) with 8+ years designing, building, and operating high-scale, production-grade cloud distributed systems and agentic AI platforms in mission-critical environments. Hands-on technical leader across architecture, code, reviews, and incident response, with deep expertise in LLM orchestration (LangGraph, LangChain, MCP), tool calling, RAG, memory, evaluation, and guardrails. Delivered fault-tolerant, multi-tenant services on Oracle Cloud Infrastructure (OCI) at 99.99% availability with sub-second latency for 1K+ daily users, a 35% production accuracy lift, and SLI/SLO and error-budget driven reliability programs.

KEY SKILLS

Languages & Frameworks: Java (Spring Boot), GoLang (goroutines, channels, worker pools), Python (FastAPI), C#, TypeScript, SQL; production code quality, code reviews, testing (JUnit, Pytest), debugging complex distributed environments

Agentic AI & LLM Systems: Agentic AI platforms, LLM orchestration, LangGraph, LangChain, MCP (Model Context Protocol), function/tool calling, multi-step task orchestration, reasoning, planning, human-in-the-loop escalation, RAG, embeddings, vector search (pgvector), prompt design, structured outputs, context management, memory, tool safety, guardrails, evaluation; AI-assisted development (Claude Code, Cursor, Copilot)

Distributed Systems & Cloud: High-scale distributed systems, microservices, horizontal and vertical scaling, distributed state management (Redis), redundancy, replication, automated failover, partition tolerance, multi-tenant cloud services, Kafka, event-driven queues, REST APIs, service-to-service communication, Kubernetes, Docker, cloud-native infrastructure, OCI (Compute, OKE, Object Storage, VCN, Load Balancers, Monitoring), AWS, GCP, high-throughput and low-latency design

Reliability & Operations: SLIs/SLOs, error budgets, production readiness, telemetry, KPIs, dashboards, alerts, tracing, observability, incident response, root cause analysis, runbooks, change management, deployment safety, rollbacks, load testing, fault injection, brownout and controlled-degradation testing, bulkheads, request shedding, Infrastructure as Code (Terraform), CI/CD, on-call livesite operations

Security & Data: OCI IAM policies, compartments, dynamic groups, instance principals, least-privilege access, multi-tenant access controls, encryption in transit and at rest, security gap remediation, secrets management; PostgreSQL, MongoDB, DynamoDB, Redis, Snowflake

Leadership & Process: Technical leadership without direct authority, mentoring, design reviews, coding standards, software effort estimation, Agile/Scrum, technical strategy, cross-team collaboration, stakeholder communication, technical documentation

PROFESSIONAL EXPERIENCE

Sprouts AI | Software Engineer (Principal, Agentic AI Infrastructure) | Chicago, IL

Mar 2025 - Present

- Increased platform adoption to 1K+ daily users at sub-second latency by architecting a scalable agentic AI system in Python/FastAPI capable of reasoning, planning, tool use, multi-step task orchestration, and safe human-in-the-loop escalation, using LangGraph, MCP, and OpenAI function calling.
- Improved production recommendation accuracy 35% by building RAG and embedding-based retrieval on PostgreSQL (pgvector) with prompt design, structured outputs, context management, guardrails, and automated LLM evaluation pipelines.
- Sustained 99.99% availability for mission-critical tier-one services by defining SLIs, SLOs, and error budgets and implementing graceful degradation with timeouts, retries with exponential backoff and jitter, circuit breakers, bulkheads, request shedding, idempotency, and fallback workflows, reducing blast radius of downstream failures.
- Deployed high-throughput GoLang and Python microservices on Oracle Kubernetes Engine (OKE) across OCI Compute, IAM, Object Storage, VCNs, and Load Balancers, with node pools, autoscaling, rolling deployments, health probes, workload isolation, and instance-principal least-privilege access.
- Prevented production outages by driving brownout and controlled-degradation testing (dependency latency injection, traffic throttling, partial-failure simulation) in release readiness, with dashboards and alerts on error-budget burn rate, p95/p99 latency, saturation, and recovery time.

Resilience Inc | Software Engineer (Distributed Systems, Cloud Services) | Chicago, IL

Aug 2023 - Mar 2025

- Cut page load time 30% for 10K+ users by architecting stateless, horizontally scalable microservices with Kafka event-driven communication, distributed state management on Redis, and REST APIs on cloud-native infrastructure.
- Eliminated state corruption under network unreliability, partitions, and duplicate events by designing services and background workers with idempotency, deduplication, rate limiting, and backpressure while consistently meeting SLOs.
- Doubled release frequency with zero-downtime deploys by building CI/CD automation (GitHub Actions) and Terraform Infrastructure as Code with reusable modules across dev, staging, and production, plus containerized Kubernetes deployments and safe rollback tooling.
- Reduced mean time to resolution 40% by establishing observability with telemetry, distributed tracing, dashboards, and alerting, owning on-call incident response and root cause analysis, and storing durable artifacts, logs, and backups in object storage with lifecycle policies.

Tata Steel | Software Engineer (Java, Core Cloud Infrastructure) |

Apr 2018 - Aug 2023

- Improved API response times 40% at 99.99% uptime by architecting cloud-native Java (Spring Boot) and C#/.NET microservices on large-scale, highly distributed infrastructure with horizontal scaling and high availability at enterprise scale.
- Hardened multi-tenant enterprise platforms by implementing secure infrastructure controls including role-based access controls, encryption in transit and at rest, and remediation of security gaps in a security-sensitive environment.
- Reduced manual processing time 60% by delivering integration-heavy enterprise platforms (Learning & Development, IRIS, Coaching) end to end across the full SDLC, applying SOLID principles, dependency injection, and async patterns.
- Strengthened operational readiness of tier-one services by leading deployment monitoring, incident triage, root cause analysis, change management, and runbook and recovery process improvements with failure-domain isolation.
- Raised the engineering bar by providing technical leadership to a cross-functional team, mentoring engineers, reviewing designs, providing software effort and complexity estimates for planning, and establishing Agile processes, coding standards, and CI/CD pipelines.

PROJECTS

Agentstore | Multi-Agent Orchestration Platform (LangGraph, MCP, Python) | [Source Code](#) | [MCP](#)

- Built an agentic AI platform with LangGraph supervisor-worker orchestration, tool calling, memory, MCP server integration, and evaluation hooks for multi-step autonomous workflows.

Voting App | Cloud-Native Microservices on Kubernetes (Docker, K8s) | [Source Code](#)

- Deployed a multi-service distributed application on Kubernetes with Docker, health probes, horizontal autoscaling, rolling in-service upgrades, and service-to-service communication.

PUBLICATIONS

IEEE: [Comparative Study of Movie Recommendation Systems using Feature Engineering](#) ([Source Code](#), [Certificate](#))

EDUCATION

